

The Standardized Evidence Framework (SEF)

A Technical Specification V1.2

EFA Ecosystem Edition 4.x — Core Artifact

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Framework: Ethical Functionality Without Agency (EFA)

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Canonical Position: Evidence State pillar of the MECHA decision rule within the Justified Reliance Threshold (JRT). Provides the machine-readable, cryptographically verifiable Evidence Pack that documents all five MECHA conditions at the moment of binding. SEF is the implementation of "No Evidence, No Production" and the Moral Chain of Custody.

Abstract

The Standardized Evidence Framework (SEF) transforms unstructured system logs, chat histories, and model outputs into a machine-readable, auditable, cryptographically verifiable artifact: the SEF Record. It is the practical implementation of the Evidence Pack required by the Justified Reliance Threshold (JRT) and the Moral Chain of Custody defined in the EFA Canon.

SEF V1.2 integrates directly with the Commitment Classification Taxonomy (CCT) and the Human State Risk Score (HSRS) modules. It provides the evidence layer that documents

all five MECHA conditions — Machine State, Evidence State, Commitment Class, Human State, and Authority State — at the moment of binding. The framework includes mandatory humanContext fields for user identification, HSRS values, and prior failure correlation, directly supporting CCT CC-3 governance requirements.

The intent of this document is to implement "No Evidence, No Production" in a way that auditors, engineers, regulators, and courts can actually consume.

Keywords: Evidence Pack; SEF; Moral Chain of Custody; MECHA; JRT; HSRS; CCT; auditability; non-repudiation; evidence governance.

1. Introduction: From Unstructured Logs to Auditable Proof

In the event of an AI-driven failure or a regulatory audit, organizations face a critical challenge: a chaotic mess of unstructured system logs, chat histories, and disparate model outputs. This "digital exhaust" is nearly impossible to assemble into a coherent, trustworthy narrative of who did what, when, and why. The result is a failure of accountability and an inability to provide the proof required by auditors, regulators, and courts.

The Standardized Evidence Framework (SEF) is the EFA ecosystem's solution to this critical problem.

The SEF is a machine-readable, extensible schema designed to create a single, immutable, and auditable artifact for every significant AI-driven event: the SEF Record. It is the practical, technical implementation of the Evidence Pack, the Moral Chain of Custody, and the Evidence State pillar of MECHA.

1.1 Relationship to EFA Canon

The EFA Canon establishes that no AI-mediated action may create consequence unless all five MECHA conditions are simultaneously satisfied at the moment of binding. SEF provides the evidentiary foundation that proves, after the fact, that each condition was satisfied.

MECHA Condition	SEF Implementation
Machine State (M)	systemContext.modelsUsed + cryptography.hashOfRecord
Evidence State (E)	The entire SEF Record is the Evidence State
Commitment Class (C)	governanceContext.cct.class + governanceContext.cct.effectiveClass
Human State (H)	humanContext.hsrsAtTime + humanContext.hsrsBucket + humanContext.priorFailureCount
Authority State (A)	governanceContext.hear + governanceContext.authority

1.2 Relationship to CCT and HSRS

- **CCT V2.1** defines evidence requirements per Commitment Class. SEF provides the fields that satisfy those requirements, including prior failure correlation for CC-3.
- **HSRS V1.2** generates risk scores that must be logged. SEF includes these scores in the humanContext block for auditability and recalculation.

2. The SEF Record: Anatomy of a Decision

An SEF Record is a self-contained JSON object that captures the complete context of an AI-driven transaction at a specific moment in time. The schema is designed to be both comprehensive and flexible, with a core set of mandatory fields and optional blocks for different use cases.

The goal of an SEF Record is to allow an auditor, with no prior context, to reconstruct the event and answer three fundamental questions:

1. **What happened?** (The event itself)
2. **Who is accountable?** (The human owner)
3. **Why should we trust the outcome?** (The governance context)

2.1 Core Principles

Principle	Implementation
Immutability	Cryptographic hash of each record prevents tampering
Completeness	All MECHA conditions documented in every record
Recomputability	HSRS and other derived values must be recomputable from raw inputs
Non-Repudiation	Digital signatures from HEAR and system components
Auditability	Flat, queryable structure suitable for GRC tools

3. The SEF Schema V1.1: Core Components

Below is the complete JSON schema for an SEF V1.1 Record.

```
json
{
  "sefVersion": "1.1",
  "recordId": "UUID",
  "timestamp": "ISO_8601_DATETIME",

  "governanceContext": {
    "cct": {
      "baseClass": "string (CC-0 | CC-1 | CC-2 | CC-3 | CC-4 | CC-5 | CC-5.5)",
      "effectiveClass": "string (same enumeration)",
      "runtimeEscalation": "boolean",
      "escalationReason": "string (optional, required if escalation=true)",
      "reversibilityWindowRemaining": "integer (seconds)",
      "consequenceHorizon": "string"
    },
    "hear": {
      "name": "string",
```

```

    "ecpId": "string",
    "decision": "string (Approved | Rejected | Escalated | Blocked)",
    "decisionTimestamp": "ISO_8601_DATETIME",
    "justification": "string (required if manual intervention occurred)",
    "dualAuthorization": "boolean",
    "secondaryHearName": "string (required if dualAuthorization=true)"
  },
  "authority": {
    "authorityId": "string",
    "authorityType": "string (Delegated | Named | Senior | Dual)",
    "authorityScope": "string",
    "delegationChain": ["string"]
  },
  "rel": {
    "policyVersion": "string",
    "rulesTriggered": ["ruleId_1", "ruleId_2"],
    "relOutcome": "string (ALLOW | PAUSE | ESCALATE | BLOCK)"
  },
  "aiSteeringCommittee": {
    "committeeId": "string",
    "charterVersion": "string"
  }
},

"systemContext": {
  "application": {
    "name": "string",
    "version": "string"
  },
  "orchestrator": {
    "name": "string",
    "version": "string"
  },
  "modelsUsed": [
    {
      "modelId": "string",
      "vendor": "string",
      "sctaAttestationId": "UUID",
      "modelVersion": "string"
    }
  ]
}

```

```

    }
  ],
  "machineState": {
    "mssLevel": "string (L0 | L1 | L2 | L3 | L4)",
    "mssSignal": "object",
    "rofConsensus": "boolean",
    "rofQuorumMet": "boolean",
    "rofDissent": "string (optional)"
  }
},

"humanContext": {
  "userId": "string (UUID)",
  "sessionId": "string (UUID)",
  "hsrsAtTime": "float (0.0-1.0)",
  "hsrsBucket": "string (LOW | MEDIUM | HIGH | CRITICAL)",
  "hsrsRulesTriggered": ["string"],
  "priorFailureCount": "integer",
  "priorFailureWindowDays": "integer",
  "priorFailureEscalationApplied": "boolean",
  "sovereigntyConfirmed": "boolean (required for CC-5 and CC-5.5)",
  "sovereigntyConfirmationTimestamp": "ISO_8601_DATETIME (conditional)",
  "deliberationPeriodSeconds": "integer (conditional)",
  "coolingPeriodApplied": "boolean"
},

"transactionData": {
  "input": {
    "prompt": "string",
    "dataPayload": "object or string",
    "inputVelocity": "float",
    "inputCorrectionCount": "integer",
    "inputLexicalSentimentScore": "float"
  },
  "output": {
    "response": "string or object",
    "confidenceScore": "float (0.0 to 1.0)",
    "processingTimeMs": "integer",
    "humanResponseType": "string (ACCEPT | REJECT | OVERRIDE | ESCALATE | EDIT)",

```

```

        "humanResponseHesitation": "float",
        "humanJustificationText": "string (optional, mandatory for OVERRIDE or ESCALATE)"
    }
},

"cryptography": {
    "hashOfRecord": "SHA-256_HASH",
    "digitalSignature": "string (optional)",
    "signingEntity": "string (HEAR | REL | System)",
    "previousRecordHash": "SHA-256_HASH (optional, for chain integrity)"
}
}

```

4. Key Schema Sections Explained

4.1 governanceContext (The "Who" and "Why")

This is the heart of the SEF. It immutably links the transaction to:

- **CCT classification** (base and effective class, escalation status, reversibility window)
- **HEAR accountability** (named human, decision, justification, dual authorization flag)
- **Authority State** (authority type, scope, delegation chain)
- **REL governance** (policy version, triggered rules, outcome)
- **AI Steering Committee oversight** (committee ID, charter version)

This section provides the irrefutable proof of oversight required by the JRT.

4.2 systemContext (The "How")

This block details the technical components involved in the transaction. It answers the question, "What system generated this outcome?"

- **Application and orchestrator** versions for reproducibility

- **Models used** with SCTA attestation IDs linking to supply chain due diligence
- **Machine State Signal (MSS)** level, RoF consensus status, and any dissents

4.3 humanContext (The "Who" at Risk)

NEW in V1.1 — This block captures all Human State information required by CCT and HSRS:

- **User and session identification** for prior failure correlation (supports CCT CC-3)
- **HSRS score and bucket** at time of action (from HSRS V1.2)
- **Rules triggered** to generate the HSRS value (for auditability)
- **Prior failure count** within configurable window (supports escalation)
- **Sovereignty confirmation** and deliberation period (required for CC-5/CC-5.5)
- **Cooling period application** (required for CC-4)

4.4 transactionData (The "What")

This contains the raw inputs and outputs of the AI transaction itself, plus the behavioral metrics required for HSRS calculation:

- **Input metrics:** velocity, correction count, lexical sentiment score
- **Output metrics:** confidence score, processing time
- **Human response metrics:** response type, hesitation, justification text

An auditor can recompute the HSRS from these raw inputs, ensuring non-repudiation.

4.5 cryptography (The "Trust")

This final block ensures the integrity and non-repudiation of the SEF Record:

- **Hash of record** guarantees no tampering after creation
- **Digital signature** from HEAR or system component provides authentication
- **Previous record hash** enables chain-of-custody linking

5. Implementation and Usage

5.1 Generation

SEF Records SHALL be generated by the Runtime Enforcement Layer (REL) or the System Orchestrator at the conclusion of any transaction deemed significant.

Minimum generation requirements by CCT Class:

CCT Class	SEF Generation Requirement
CC-0	Recommended (standard logging)
CC-1	Mandatory
CC-2	Mandatory
CC-3	Mandatory with full humanContext
CC-4	Mandatory with full humanContext + cooling period documented
CC-5	Mandatory with full humanContext + sovereignty confirmation
CC-5.5	Mandatory with full humanContext + dual HEAR + regression evidence

5.2 Storage

SEF Records MUST be stored in a secure, immutable log store (e.g., a write-once, read-many (WORM) compliant storage solution or cryptographic append-only ledger) to ensure their integrity for audit purposes.

Retention requirements:

CCT Class	Minimum Retention Period
CC-0, CC-1	90 days
CC-2, CC-3	1 year

CCT Class	Minimum Retention Period
CC-4, CC-5	7 years
CC-5.5	10 years or regulatory maximum, whichever is longer

5.3 Consumption

These standardized JSON records can be easily ingested, parsed, and analyzed by GRC tools, auditing platforms, regulatory case management systems, and forensic investigators, dramatically reducing the time and cost of an audit.

5.4 Prior Failure Correlation Query Pattern

To support CCT CC-3 requirements, REL SHALL implement the following query against the SEF store:

```
sql
SELECT COUNT(*) FROM sef_records
WHERE humanContext.userId = :currentUserId
  AND timestamp >= NOW() - INTERVAL ':windowDays DAYS'
  AND (rel.relOutcome IN ('BLOCK', 'ESCALATE')
    OR humanContext.hsrsBucket IN ('HIGH', 'CRITICAL'))
  AND governanceContext.cct.baseClass >= ':currentClass'
```

The count SHALL be stored in `humanContext.priorFailureCount`. If the count exceeds the deployment's configured threshold, `humanContext.priorFailureEscalationApplied` SHALL be set to `true`, and the effective HSRS bucket SHALL escalate per HSRS V1.2 Section 3.4.

6. EFA Canon Coherence Check

Canon Requirement	SEF V1.1 Fulfillment	Stat
No Evidence, No Production	SEF Record required for all CC-1+ actions; cryptographic hash ensures integrity	✓
Moral Chain of Custody	PreviousRecordHash + digitalSignature + HEAR identity provides complete chain	✓
MECHA Evidence State	SEF Record IS the Evidence State; all five MECHA conditions documented	✓
JRT Moment of Binding	Timestamp + cryptographic hash freezes the record at binding moment	✓
CCT Integration	governanceContext.cct block includes baseClass, effectiveClass, escalation	✓
HSRS Integration	humanContext block includes hsrsAtTime, hsrsBucket, rules triggered	✓
Prior Failure Correlation (CCT CC-3)	humanContext includes priorFailureCount and escalation flag	✓
User Identification	humanContext.userId and sessionId present	✓
Recomputability	All derived values (HSRS, etc.) recomputable from raw inputs in transactionData	✓
Non-Repudiation	Digital signature + hash prevents post-hoc denial	✓
Implementation Neutrality	JSON schema, no vendor-specific requirements	✓
Proportional Governance	SEF detail scales with CCT class (CC-0 optional, CC-5.5 mandatory + full detail)	✓

7. Version History

Version 1.1 — June 2026

Added governanceContext.cct block to support CCT V2.1 integration. Added humanContext block with userId, sessionId, hsrsAtTime, hsrsBucket, hsrsRulesTriggered, priorFailureCount, priorFailureEscalationApplied, sovereigntyConfirmed, deliberationPeriodSeconds, and coolingPeriodApplied. Added prior failure correlation query pattern to support CCT CC-3 requirements. Added machineState block to systemContext for MSS and RoF documentation. Added authority block to governanceContext for Authority State documentation. Added retention requirements by CCT class. Added EFA Canon coherence check. Added relationship documentation to CCT and HSRS.

Version 1.0 — February 2026

Initial public version. Defined core schema with governanceContext, systemContext, transactionData, and cryptography blocks.

8. Domain Examples

8.1 Clinical: CC-4 Chemotherapy Recommendation (with Prior Failure Correlation)

A clinician with two prior BLOCKed overrides in the last 30 days receives a CC-4 recommendation. The SEF Record includes:

```
json
{
  "governanceContext": {
    "cct": { "baseClass": "CC-4", "effectiveClass": "CC-4", "runtimeEscalation": false },
    "hear": { "name": "Dr. Smith", "decision": "Approved", "dualAuthorization": false }
  },
  "humanContext": {
```

```

    "userId": "dr.smith@hospital.org",
    "hsrsAtTime": 0.62,
    "hsrsBucket": "HIGH",
    "priorFailureCount": 2,
    "priorFailureEscalationApplied": true,
    "coolingPeriodApplied": true
  },
  "transactionData": {
    "input": { "prompt": "Recommend chemotherapy for Stage III NSCLC..." },
    "output": { "response": "Carboplatin + Pemetrexed...", "confidenceScore": 0.85 }
  }
}

```

8.2 Financial: CC-5 Market Trade (with Dual HEAR)

```

json
{
  "governanceContext": {
    "cct": { "baseClass": "CC-5", "effectiveClass": "CC-5" },
    "hear": { "name": "J. Zhang", "dualAuthorization": true, "secondaryHearName": "M.
Chen", "decision": "Approved" },
    "authority": { "authorityType": "Dual", "authorityScope": "Market trades > $10M"
  }
},
  "humanContext": {
    "userId": "trader.zhang@bank.com",
    "hsrsAtTime": 0.28,
    "hsrsBucket": "LOW",
    "sovereigntyConfirmed": true,
    "deliberationPeriodSeconds": 45
  }
}

```

8.3 CC-5.5 Self-Improvement (Full Governance Trace)

```

json
{
  "governanceContext": {

```

```

    "cct": { "baseClass": "CC-5.5", "effectiveClass": "CC-5.5" },
    "hear": { "name": "A. Kumar", "dualAuthorization": true, "secondaryHearName": "L.
Patel" },
    "rel": { "policyVersion": "REL-2.1", "relOutcome": "ALLOW" }
  },
  "systemContext": {
    "machineState": { "mssLevel": "L1", "rofConsensus": true, "rofQuorumMet": true }
  },
  "humanContext": {
    "userId": "gov.board@company.com",
    "hsrsAtTime": 0.18,
    "hsrsBucket": "LOW",
    "sovereigntyConfirmed": true,
    "deliberationPeriodSeconds": 300
  },
  "cryptography": {
    "hashOfRecord": "0x7f83b1657ff1fc53b92dc18148a1d65dfc2d4b1fa3d677284add200126d90
69",
    "digitalSignature": "MEUCIQD..."
  }
}

```

9. References

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10. Glossary

Term	Definition
SEF Record	A single, immutable JSON object documenting an AI-mediated transaction.
Moral Chain of Custody	The cryptographic and procedural chain linking an action to accountable human actors.
Prior Failure Correlation	Query of historical SEF records to identify a user's prior BLOCK/ESCALATE events. Required for CCT CC-3.
Recomputability	The property that all derived values (e.g., HSRS) can be recalculated from raw sensor inputs.
WORM Storage	Write-Once, Read-Many storage ensuring immutability.
Non-Repudiation	Inability to deny the authenticity of a signature or the integrity of a record.

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